Could you turn the sun down a bit?

"The belief that we can manage the Earth and improve on Nature is probably the ultimate expression of human conceit..."
–Rene J. Dubos

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Humans will always find a way to mould things around us to suit us, and that is why we are awesome, and that is also probably why we are hated (by hyper intelligent animals who can understand such complex emotions). Glow in the dark rabbits? Poisonous cabbage? Tosh! We'll take you through some of the more interesting, and philosophical-question-raising ideas that scientists around the world have come up with to make our earth a better place.

I've always wanted a lightning-facing view

Dial-a-lightning

In what would make a very good slow-motion video, all scientists at the University of Florida have to do is fire a rocket with a trailing grounded copper wire attached to it into a thundercloud, and after travelling around 600-900 feet --lightning strikes. The special wire inside the three foot long rocket distorts the electric field under the cloud, resulting in a lightning discharge hitting the rocker.

All of this occurs with the team behind it cosy in a metallic trailer that is completely grounded. The alternative, is to hope that lightning falls *exactly* where you want it to. The main scientist behind this, Dr. Rakov and his colleagues use this technique to test to what extent devices like electric lines, insulation on nuclear weapons, and airport runway lights can withstand such a hit.

But this is when you want lightning to hit something. There are also ways to make lightning go somewhere else. While still in the test phase, the idea is of shooting a beam into the atmosphere that will leave a stream of ionized molecules in its wake, minus their electrons. This stream could be used to pull and channel lightning bolts.

The only problem was that till now all of the research just allowed for this beam to travel a few feet at best, useless we would imagine unless you want that lightning bolt to hit "just a few feet away." Now, courtesy University of Arizona and Florida, that beam can travel at least 165 feet, enough to avert a disaster.

The new beaming method will involve embedding the primary high intensity laser beam into a secondary lower intensity beam, called the dress beam, which will refuel the primary beam and allow it to travel larger distances.

Ack! You say this was pretty straightforward. Where are the philosophical-question-raising ideas you spoke about? Yes yes we are getting to it, gradually.

Hurricane Repel

When they are not being helpful by destroying Mongol fleets when they were en-route to invade Japan 1274, or preventing the England from being attacked by the Spanish Armada in 1588 by ramming right into the invaders, hurricanes cause billions of dollars of damage by destroying homes, levelling cities, and killing thousands. Yes there are methods now which ensure that hurricanes do not arrive unannounced, but there is only so much one can do in 48 hours, shifting an entire city not being one of the options. So what can we do?

Well between 1962 and 1983, a hurricane research program called Project *Stormfury* was implemented with a plan